

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12384566
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Casari; Pascal et al.

Drone chassis

Abstract

A drone chassis including an external structure that deforms elastically in bending, and a set of cables connecting at least one support, intended to support a functional element of the drone, to the external structure. The set of cables includes elastic cables deforming elastically in traction under a tensile stress above a natural deformation threshold, the set of cables being attached at several points on the external structure, referred to as attachment points, to hold the at least one support in a stable position with respect to the external structure. The elastic cables are attached under stress by applying an initial tensile stress and maintains the initial tensile stress by attaching to the external structure. An additional tensile stress applied to the cables attached under stress causes a deformation under traction of the cables only as from a threshold that is above the natural deformation threshold of the cables.

Inventors:	Casari; Pascal (Saint-Nazaire, FR), Berthiau; Thibault (Vigneux de Bretagne, FR), Trichereau; Antoine (Nantes, FR)
Applicant:	NANTES UNIVERSITE (Nantes, FR); BEE TECH (Nantes, FR)
Family ID:	71094559
Assignee:	NANTES UNIVERSITE (Nantes, FR); BEE TECH (Nantes, FR)
Appl. No.:	17/919493
Filed (or PCT Filed):	April 14, 2021
PCT No.:	PCT/EP2021/059620
PCT Pub. No.:	WO2021/209483
PCT Pub. Date:	October 21, 2021

Prior Publication Data

Document Identifier	Publication Date
----------------------------	-------------------------

Foreign Application Priority Data

FR

2003869

Apr. 17, 2020

Publication Classification**Int. Cl.:** B64U20/73 (20230101); B64U10/14 (20230101); B64U20/70 (20230101)**U.S. Cl.:****CPC** B64U20/73 (20230101); B64U20/70 (20230101); B64U10/14 (20230101)**Field of Classification Search****CPC:** B64U (20/73); B64U (20/70); B64U (10/10); B64C (39/024)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9840339	12/2016	O'Brien	N/A	B64C 39/024
9908619	12/2017	Beckman	N/A	B64U 40/20
2016/0280359	12/2015	Semke	N/A	B64U 60/60
2016/0375983	12/2015	Yan et al.	N/A	N/A
2017/0050726	12/2016	Yamada	N/A	B64U 60/60
2017/0291697	12/2016	Kornatowski	N/A	B64C 39/024
2017/0297681	12/2016	Yamada	N/A	B64U 30/299
2017/0329034	12/2016	Miles et al.	N/A	N/A
2017/0334579	12/2016	Hartkop	N/A	B64U 10/14
2018/0057135	12/2017	Yan	N/A	B64C 1/08
2018/0224262	12/2017	Klein	N/A	B64U 20/10
2020/0277054	12/2019	Lee	N/A	B64U 10/16
2021/0155346	12/2020	Liarokapis	N/A	B64U 30/20
2021/0237897	12/2020	Gury	N/A	G05D 1/102
2022/0380044	12/2021	Arnold	N/A	B64D 47/08
2022/0396369	12/2021	Hartkop	N/A	B64U 20/87
2023/0192329	12/2022	Casari	244/121	B64U 20/73

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
98/15429	12/1997	WO	N/A
2014/198774	12/2013	WO	N/A
2018/156991	12/2017	WO	N/A

OTHER PUBLICATIONS

Jun. 18, 2021 International Search Report issued in International Patent Application No. PCT/EP2021/059620. cited by applicant

Primary Examiner: Michener; Joshua J

Assistant Examiner: Curry; Cindi M

Attorney, Agent or Firm: Williams Mullen

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to the field of drones and more particularly the field of drone chassis intended to protect the drone and the environment thereof in the event of impact and to provide stability of functional elements associated with the drone.

PRIOR ART

(2) Drones are being used for more and more numerous applications and are therefore caused to encounter varied obstacles. Consequently a drone may be subjected to impacts of variable intensity according to the flight situations and the potential obstacles present in the environment of the drone. Such impacts have the consequence of damaging the drone, whether structurally by damaging the chassis or functionally by degrading functional elements such as a flight map, a battery or motors. Impacts between the drone and external objects may furthermore cause human and physical damage in the environment of the drone.

(3) Some drones have a rigid structure that makes it possible to withstand high impact speeds by absorbing a large amount of energy by plastic deformation. However, such drones do not make it possible to limit the damage caused to their environment. Other drones with a less rigid structure withstand low impact speeds by elastic absorption of energy. On the other hand, such structures may lack stability.

(4) It is therefore desirable to overcome these drawbacks of the prior art.

(5) It is in particular desirable to provide a solution that makes it possible to protect the drone and its environment during impacts at various speeds and therefore for variable impact forces. It is furthermore desirable to provide a solution that is lightweight. It is also desirable to provide a solution that affords stable holding of the functional elements of a drone in normal flight situations and thus guarantees optimum operation of the drone.

DISCLOSURE OF THE INVENTION

(6) One object of the present invention is to propose a drone chassis comprising an external structure that deforms elastically in bending, and furthermore comprising a set of cables connecting at least one support, intended to support a functional element of the drone, to the external structure. The set of cables comprises elastic cables that deform elastically in traction under a tensile stress above a threshold, referred to as the natural tensile deformation threshold. The set of cables is attached at several points on the external structure, referred to as attachment points, to hold the at least one support in a stable position with respect to the external structure. The elastic cables are attached under stress by applying, to said elastic cables, an initial tensile stress, and maintaining the initial tensile stress by attaching to the external structure so that an external force applied to the chassis and generating an additional tensile stress on the elastic cables attached under stress causes a deformation under traction of the cables and a bending deformation of the external structure only if said additional tensile stress exceeds a minimum tensile deformation threshold that is above the natural tensile deformation threshold of the elastic cables.

(7) Thus the drone chassis withstands low impact speeds without deformation of the external structure while remaining lightweight. Functional elements of the drone can be held in a stable position during collisions at low impact speeds corresponding to normal flight situations.

Furthermore, the drone chassis withstands higher impact speeds by converting kinetic energy into elastic energy or by plasticising on the cables, which makes it possible to protect both the drone and its environment by reducing the amplitude of the impact forces.

(8) According to a particular embodiment, the set of cables comprises elastic cables made from high-density polyethylene.

(9) According to a particular embodiment, a cross section of each elastic cable in the set of cables is determined for reducing an elastic limit of the elastic cables and favouring plastic deformation of said cables when an additional tensile stress applied to the elastic cables attached under stress, and generated by the external force applied to the chassis, is above the minimum tensile deformation threshold of said elastic cables attached under stress.

(10) Thus absorption by plasticising of the kinetic energy received by the drone chassis during an impact is favoured compared with conversion into elastic energy, thus limiting transmission of the kinetic energy to the support and thus making it possible to protect the functional elements of the drone. According to a particular embodiment, the external structure of the drone chassis comprises two arches, a first arch being located in a plane orthogonal to a second arch, the first arch being in a vertical plane and the second arch being in a horizontal plane when the drone is in a stabilised flight position. The attachment points to which the at least one support is attached are located on one and the same arch.

(11) Thus an initial tensile stress can easily be applied and maintained on the elastic cables to which the support is attached.

(12) According to a particular embodiment, the set of cables comprises cables oriented in several directions.

(13) Thus the drone chassis can withstand external forces applied to the external structure in various directions and therefore at impacts with obstacles coming from various directions.

(14) According to a particular embodiment, a support comprises a flight map, and each cable in the set of cables has a given length for holding the support comprising the flight map at the centre of the external structure.

(15) Thus the flight map is held in a position favouring optimum guidance of the drone.

(16) According to a particular embodiment, each cable in the set of cables has a given length for centring, with respect to the external structure, a centre of gravity of a set of the functional elements supported by the at least one support.

(17) Thus the equilibrium of the drone chassis is promoted, which affords better control of the movements and therefore of the guidance of the drone.

(18) According to a particular embodiment, the set of cables connects a plurality of supports to the external structure, each support being connected to the external structure by a subset of the set of cables independently of another support.

(19) Thus it is possible to limit the transmission of a movement of a functional element, caused by a conversion into elastic energy on a subset of cables, to another functional element.

(20) According to a particular embodiment, the set of cables comprises a subset of cables connecting the at least one support to two attachment points of the external structure, the subset of cables comprising two pairs of cables, each pair of cables connecting one of the two attachment points to the support, and the subset of cables being arranged so that, for each pair of cables, a first cable in the pair of cables connects an attachment point to a first point of the support and a second cable in said pair of cables connects the same attachment point to a second point of the support different from the first point of the support.

(21) Thus an initial tensile stress can easily be applied and maintained on the elastic cables to which the support is attached.

(22) According to a particular embodiment, the set of cables comprises a subset of cables that connects, between two attachment points, several supports aligned with each other and aligned with said two attachment points.

- (23) Thus it is easy to adjust the initial tensile stress to each cable in the subset of cables when the subset of cables is attached to the second attachment point.
- (24) According to a particular embodiment, the set of cables connects the at least one support to six attachment supports of the second arch by six distinct cables, the six distinct cables comprising three upper cables connecting the support to respectively three first attachment points, the first attachment points being located at the upper part of the second arch and distributed over the circumference of the arch, and three lower cables connecting the support to respectively three second attachment points, each second attachment point being located on the second arch below a first attachment point.
- (25) According to a particular embodiment, the drone chassis supports at least four motor units each comprising a motor and a propeller, each motor unit being attached to the external structure by a rigid rod, and wherein the set of cables comprises a cable connecting each motor unit to each other motor unit.
- (26) Thus the external structure is stiffened by means of the cables connecting the motor units, which makes it possible to avoid deformation during impact at low speeds, but keeps deformability during impacts at high speeds making it possible to reduce the impact force both on the drone and on its environment.
- (27) According to a particular embodiment, electric wires are passed inside cables in the set of cables for electrically connecting functional elements together.
- (28) Thus the functional elements can be electrically connected together by means of a shortest possible routing of the electric wires, thus making it possible to limit an addition of mass to the drone chassis.
- (29) The invention also relates to a drone comprising a drone chassis mentioned above in any one of the embodiments thereof.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The features of the invention mentioned above, as well as others, will emerge more clearly from the reading of the following description of at least one example embodiment, said description being made in relation to the accompanying drawings, among which:
- (2) FIG. 1 illustrates schematically a drone according to a particular embodiment;
- (3) FIG. 2A illustrates schematically a part of a drone chassis with a set of cables when the cables are not completely attached to an external structure.
- (4) FIG. 2B illustrates schematically a part of the drone chassis with a set of cables when the cables are completely attached to the external structure.
- (5) FIG. 3 illustrates schematically a view in cross section, along a first cutting plane, of a part of the drone chassis, the chassis comprising a set of cables for holding a support according to a first embodiment;
- (6) FIG. 4 illustrates schematically a view in cross section, along a second cutting plane, of a part of the drone chassis, the chassis comprising a set of cables for holding a plurality of supports according to a second embodiment;
- (7) FIG. 5 illustrates schematically a view in cross section, along the first cutting plane, of a part of the drone chassis, the chassis comprising a set of cables for holding a plurality of supports according to a third embodiment;
- (8) FIG. 6 illustrates schematically a view in cross section, along the first cutting plane, of a part of the drone chassis, the chassis comprising a set of cables for holding a plurality of supports according to a fourth embodiment;
- (9) FIG. 7 illustrates schematically in plan view a part of the drone chassis, with the set of cables of

the drone chassis connecting four motors each connected to a propeller;

(10) FIG. 8 illustrates schematically in perspective a part of the drone chassis with a set of cables making it possible to hold a support according to a fifth embodiment.

DETAILED DISCLOSURE OF EMBODIMENTS

(11) FIG. 1 thus illustrates schematically a drone **100** according to a particular embodiment. The drone **100** comprises a chassis comprising an external structure **110**.

(12) The external structure **110** is for example in the form of a cage. In the embodiment shown in FIG. 1, the external structure **110** comprises a first arch **111** and a second arch **112**, protective crescents **114** and wires **113** for connecting the arches **111**, **112** and the protective crescents **114** to each other. The first arch **111** is located in a plane orthogonal to the second arch **112**. When the drone **100** is in a stabilised flight position as shown in FIG. 1, the first arch **111** is in a vertical plane and the second arch **112** is in a horizontal plane.

(13) The external structure **110** is elastic under bending. In other words, the external structure **110** deforms reversibly when it is subjected to a bending stress that exceeds a first predefined threshold, referred to as the minimum bending deformation threshold of the external structure **110**, and regains its repose shape when the bending stress once again goes below said minimum bending deformation threshold of the external structure **110**. When the external structure **110** is subjected to a bending stress that exceeds a second predefined threshold referred to as the bending elastic limit of the external structure **110**, said external structure **110** deforms plastically, in other words irreversibly, and does not regain its repose shape when the bending stress once again goes below the minimum bending deformation threshold of the external structure **110**. The bending elasticity of the external structure **110**, in the event of impact, causes bending deformations that are elastic and therefore reversible. Furthermore, the bending elastic deformations of the external structure **110** caused by an impact extend a duration of contact between the external structure **110** and an obstacle. An impact force is thus distributed over a longer period of contact, which causes a reduction in the amplitude of the impact force. The bending elastic deformations of the external structure **110** moreover cause an increase in a contact surface area between said external structure **110** and the obstacle encountered. The energy transmitted during the impact is then distributed over a larger surface area, causing a local reduction in a force peak and therefore a reduction in a stress applied by the impact force, both on the drone and on an obstacle.

(14) The ability of the external structure **110** to deform elastically under bending thus makes it possible, by favouring reversible deformations and reducing the bending stresses applied to the external structure **110**, on the one hand to limit damage to the external structure **110** and on the other hand to limit any damage that may be caused to obstacles in the environment of the drone such as objects or persons.

(15) The external structure **110** comprises elements produced with lighter composite materials, having a higher bending elastic limit and thus making it possible to produce thinner parts than metallic materials for example. The external structure **110** thus has limited mass. Obtaining thinner parts, thus comprising less material, furthermore makes it possible to increase the bending deformability of the external structure **110**.

(16) An element of the external structure **110** such as an arch **111**, **112**, a protective crescent **114** or a wire **113** is advantageously curved in order to avoid zones liable to concentrate stresses. Furthermore, curved shapes make it possible to limit the mass of the external structure **110** and the quantity of material used. Thus the external structure **110** may for example have an ellipsoidal shape or a trefoil shape.

(17) The drone chassis **100** furthermore comprises a set of cables **120** for holding one or more functional elements inside the internal structure **110**. A functional element is for example a flight map, a battery, one or more motors with propeller, a camera or an effector such as a sensor intended to make measurements, a product to be deposited, a parcel to be transported or a gripper dedicated to sampling. The set of cables **120** connects at least one support **130** intended to support a

functional element on the external structure **110**. A support **130** is for example a housing or a securing plate. A cable **120** may alternatively be a flexible bar, a wire, a textile link or a rope. (18) The set of cables **120** is arranged for holding the support **130** in a stable position with respect to the external structure **110**, whatever the orientation of the drone chassis **100** and as long as the external structure **110** remains in an initial geometric configuration. The set of cables **120** is attached at several points of the external structure **110**, referred to as attachment points **116**. The cables **120** are attached to the attachment points by rope connections such as knots, splices, inserts or thimbles.

(19) At least some of the cables **120** in the set of cables **120** is elastic under traction. An elastic cable **120** deforms reversibly by extending when it is subjected to a tensile stress that exceeds a third predefined threshold referred to as the natural tensile deformation of the elastic cable **120** and regains its repose shape when the tensile stress once again goes below the natural deformation threshold of the elastic cable **120**. When the elastic cable **120** is subjected to a tensile stress that exceeds a fourth predefined threshold, referred to as the tensile elastic limit of said elastic cable **120**, the elastic cable **120** deforms plastically and does not regain its repose shape when the bending stress once again goes below the natural tensile deformation threshold of the elastic cable **120**. An elastic cable **120** is for example a high-density polyethylene rope. According to other examples, an elastic cable **120** is a rope made from poly(p-phenylene-2,6-benzobisoxazole) or PBO, from carbon, from polyester, from polyamide, from latex or from polyurethane.

(20) The elastic cables **120** in the set of cables **120** are furthermore attached under stress to the external structure **110**. A so-called initial tensile stress is applied to said elastic cables **120** and maintained by attaching to the external structure **110**. In other words, an elastic cable **120** or a subset of elastic cables **120** is attached under tension between at least two attachment points **116** of the external structure **110** so that said elastic cable **120** or set of elastic cables **120** remains elastically stretched as long as the external structure **110** remains in its initial geometric configuration. The initial geometric configuration of the external structure **110** corresponds to a geometric configuration in which the set of cables **120** is totally attached to the external structure **110**, the elastic cables **120** being stressed under traction, and for which no external force is applied to the drone chassis **100**. The initial geometric configuration of the external structure **110** is therefore for example obtained in a stable flight situation. The elastic cables **120** attached under stress in the set of cables **120** make it possible, under identified flight situations, to hold the support **130** in a stable position with respect to the external structure **110**.

(21) Electric wires can furthermore be passed inside the cables **120** to electrically connect various functional elements to each other and as short as possible. For example, the motors can be electrically connected to the flight map and to the battery while limiting the mass of the drone **100**.

(22) FIG. 2A illustrates schematically a part of the drone chassis **100** with a set of cables **120** when the cables **120** are not completely attached to an external structure **110**. The part of the drone chassis **100** thus shown comprises the second arch **112** of the external structure **110** and two electric cables **120** in the set of cables **120** in a configuration where said elastic cables **120** are partially attached to the external structure **110**. In said configuration, the external structure **110** is subjected to a zero stress and is therefore in its repose shape, different from a geometric shape that the external structure **110** of the drone chassis **100** adopts when the drone **100** is in a stabilised flight situation.

(23) In order to apply an initial tensile stress to each elastic cable **120**, a length l of a cable **120** is less than a distance d between two attachment points **116** that the cable **120** is supposed to connect. The length l of a cable **120** is defined under zero deformation, in other words when the cable **120** is at repose or is not elastically stretched, and between two ends supposed to connect said two attachment points **116**. The distance d between said two attachment points **116** is a defined minimum distance under zero stress, when the external structure **110** is in its repose shape and is not attached to the set of cables **120**.

(24) For example, the cable **120a** has a length under zero deformation l that is less than the distance d under zero stress between two attachment points **116a** and **116b** that said cable **120a** is supposed to connect after attachment.

(25) FIG. 2B illustrates schematically a part of the drone chassis **100** with a set of cables **120** when the cables **120** are completely attached to the external structure. The part of the drone chassis **100** thus shown comprises the second arch **112** of the external structure **110** and two elastic cables **120** in the set of cables **120** in a configuration where said cables **120** are completely attached to the external structure **110**. The external structure **110** is then in its initial geometric configuration, which corresponds to the geometric shape that the external structure **110** of the drone chassis **100** adopts when the drone **100** is in a stabilised flight situation and which differs slightly from the repose shape of the external structure **110**.

(26) When the elastic cable **120** of length l under zero deformation is attached to the external structure **110**, between two attachment points separated by the distance d under zero stress, greater than said length l , a first effect is to apply an initial tensile stress to the elastic cable **120** that causes an elastic stretching of said elastic cable **120**. The elastic cable **120** is thus attached under stress and kept elastically stretched in the initial geometric configuration of the external structure **110**. A second effect is elastically deforming under bending at least a part of the external structure **110**, such as for example the second arch **112**, and giving to the external structure **110** its initial geometric configuration.

(27) When the drone **100** is used, the chassis may be subjected to an external force that is applied to the external structure **110**. If the external force is oriented in a direction transverse to that of a cable **120** such as for example an external force in a direction **201**, the cable **120** in question is stressed under traction and therefore subjected to an additional tensile stress. Because of the initial tensile stress applied, an elastic cable **120** attached under stress then has a behaviour that is modified compared with an elastic cable **120** that is similar but that is not attached under initial tensile stress and subjected to a tensile stress identical to said additional tensile stress. Firstly, the elongation of the elastic cable **120** attached under stress generated by the additional tensile stress is limited and therefore lower in amplitude. Moreover, the elastic cable **120** attached under initial tensile stress begins to deform when the additional tensile stress to which it is subjected is above a minimum tensile deformation threshold, said minimum tensile deformation threshold being above the natural tensile deformation threshold of the elastic cable **120**.

(28) The initial tensile stress of the elastic cables **120** thus makes it possible to limit the bending elastic deformation of the external structure **110** and to limit the movement of the support **130**. In other words, the initial tensile stress of the cables **120** makes it possible to increase the stiffness of the external structure **110**.

(29) Furthermore, as long as the additional tensile stress applied to an elastic cable **120** attached under stress is below the minimum tensile deformation threshold, said elastic cable **120** keeps an initial length. It is then possible to maintain the support **130** in a stable position with respect to the external structure **110**, which guarantees optimum operation of the functional elements supported by the support **130**. In addition, the initial geometric configuration of the external structure **110** can be kept when the chassis is subjected to an external force below a predefined fifth threshold. The predefined fifth threshold is dependent on the first and second predefined bending stress thresholds of the external structure **110**, and on the orientation of the external force with respect to the orientation of the elastic cables **120** attached under stress in the set of cables **120**, and increases with the initial tensile stress applied to the elastic cables **120** attached under stress.

(30) The initial tensile stress applied to an elastic cable **120** in the set of cables **120** can be adjusted by modifying the parameters of the cable **120** such as the length l under zero deformation of the cable **120**, the cross section under zero deformation of the cable **120** and/or the elastic limit of the cable **120**. It is thus possible to adapt the minimum tensile deformation threshold of the cables **120** by taking account of potential external forces that may be applied to the external structure **110** in

flight situations identified as normal, such as for example during landing or takeoff. Furthermore, geometric parameters of the drone chassis **100** can be adjusted by taking account of the directions of the potential external forces. In a particular embodiment, the elastic cables **120** attached under stress are oriented in several directions so as to withstand external forces applied to the external structure **110** in various directions. Thus the initial geometric configuration of the external structure **110** can be preserved in normal flight situations.

(31) FIG. 3 illustrates schematically a view in cross section, along a first cutting plane, of a part of the drone chassis comprising the first vertical arch **111** and the set of cables **120** arranged according to a first embodiment. The first cutting plane is located in the mid-plane of the drone chassis **100** and along a vertical plane when the drone **100** is in a stabilised flight position. The set of cables **120** comprises eight cables **120** each connecting the support **130** located at the centre of the first arch **111** to an attachment point **116** of the first arch **111**. The attachment points **116** are regularly distributed over the first arch **111** so that their barycentre is located at the centre of the first arch **111**. The cables **120** are grouped together in pairs. The cables **120** in each pair are aligned and oriented in the same direction and each connect the support **130** to an attachment point **116**. Each pair of cables **120** thus connects two opposite attachment points **116** on the first arch **111**.

(32) In an alternative embodiment, not shown, the set of cables **120** may comprise an odd number of cables **120**, each cable connecting an attachment point **116** of the first arch **111** to the support **130**.

(33) The attachment points **116** of the set of cables are regularly distributed over the first arch **111** so that their barycentre is located at the centre of the first arch **111**. Thus the distribution in space of the various orientations of the cables **120** makes it possible to oppose, in an equivalent manner, external forces applied with various directions.

(34) The set of cables **120** may, according to other embodiments, be in the form of a lattice, net or trellis.

(35) In the event of lateral impact on the chassis, an external force applied by an obstacle on one side of the first arch **111** generates an additional tensile stress on the elastic cables **120** attached under stress. Said elastic cables **120** attached under stress then oppose the bending deformation of the external structure **10**. If, for each elastic cable **120** attached under stress in the set of cables **120**, the additional tensile stress is below the minimum tensile deformation threshold of the cable **120**, the support **130** is maintained in a stable position with respect to the external structure **110**, and the external structure **110** keeps its initial geometric configuration.

(36) If, for at least one cable **120** in the set of cables **120**, the additional tensile stress is above the minimum tensile deformation threshold of the cable **120**, the at least one cable **120** is stretched elastically. The stability of the support **130** and the initial geometric configuration of the external structure **110** are not preserved. On the other hand, part of the kinetic energy transmitted to the chassis is converted into elastic energy. The impact force suffered by the chassis is reduced at the support **130** by spring effect because of the elasticity of said cable **120**. It is possible to adjust the stiffness of the cables **120** in order to limit the deformation of the support **130** in the event of impact.

(37) If, for at least one cable **120** in the set of cables **120**, the additional tensile stress is above the minimum tensile deformation threshold and above the elastic limit of the cable **120**, the at least one cable **120** furthermore suffers plastic deformation. Part of the kinetic energy transmitted to the chassis is thus absorbed by plasticising of the at least one cable **120** and is then not transmitted to the support **130**. Said support **130** is thus protected from the impacts. In such a case, the at least one cable **120** is damaged and must be replaced for further use of the drone **100**.

(38) The set of cables **120** can be sized to reduce the elastic limit of the elastic cables **120**, for example by reducing the cross section of the elastic cables **120**, in order to reduce an interval between the minimum tensile deformation threshold of the elastic cables **120** attached under stress and the elastic limit of the elastic cables **120**. Thus, when an external force applied has an

excessively high amplitude for the support **130** to be maintained in its initial geometric configuration by virtue of the initial tensile stress of the cables **120**, the case where the kinetic energy transmitted to the chassis is partially absorbed by plasticising is favoured.

(39) FIG. 4 illustrates schematically a view in cross section, along a second cutting plane, of a part of the drone chassis comprising the second arch **112** of the external structure **110** and the set of cables **120** arranged in a second embodiment. The second cutting plane is located in the mid-plane of the drone chassis **100** and along a horizontal plane when the drone **100** is in a stabilised flight position.

(40) The set of cables **120** comprises a first cable connecting two opposite attachment points **116** on the second arch **112**, thus forming a diameter of the second arch **112**. The set of cables **120** furthermore comprises a subset of cables **120** aligned with each other, in a direction perpendicular to the direction of the first cable. A cable **120** in the subset of cables **120** connects a first attachment point **116** of the second arch **112** to a first support **130a**, a cable **120** in the subset of cables **120** connects the first support **130a** to a second support **130b**, and a cable in the subset of cables connects the second support **130b** to a second attachment point **116** opposite to the first attachment point **116**.

(41) Said subset of cables **120** can thus connect, in one and the same alignment, two distinct functional elements such as the flight map and the battery. The cables **120** in the subset of cables can be sized to adjust the position of the functional elements supported by the supports **130a** and **130b** according to their respective masses. Thus it is for example possible to position, at the centre of the second arch **112**, the centre of gravity of the set of functional elements so as to balance the drone **100**.

(42) FIG. 5 illustrates schematically a view in cross section, along the first cutting plane, of a part of the drone chassis comprising the first vertical arch **111** of the external structure **110** and the set of cables **120** arranged according to a third embodiment. The set of cables **120** comprises two subsets of cables **120**. Each support **130a** (and respectively **130b**) is connected to the first arch **111** by a subset of cables **120** independently of another support **130b** (and respectively **130b**).

(43) Each support **130a**, **130b** is connected to two attachment points **116**, a first attachment point **116**, referred to as the upper attachment point **116**, being located above and vertically in line with a second attachment point **116**, referred to as the lower attachment point **116**. Thus the support **130a**, **130b** is held between an upper attachment point **116**, located above, and a lower attachment point **116**, located below.

(44) Each support **130a**, **130b** is furthermore connected to each of its attachment points **116** by a pair of cables **120**. A first cable **120** in each pair of cables **120** connects a first point of the support **130a**, **130b** to an attachment point **116**, and a second cable **120** in said pair of cables **120** connects a second point of the support **130a**, **130b** to the same attachment point **116**. Thus said attachment point **116** and the first and second points of the support **130a**, **130b** form a triangle.

(45) In the example of a thin first support **130a**, the upper pair of cables **120** is connected to first and second points of the second support **130b** that are identical to the first and second points on the second support **130b** to which the lower pair of cables is connected. In the example of a thicker second support **130b**, the upper pair of cables **120** is connected to first and second points on the second support **130b** that are different from the first and second points on the second support **130b** to which the lower pair of cables is connected.

(46) The two attachment points **116** to which the first support **130a** is connected are distinct from the two attachment points **116** to which the second support **130b** is connected. Thus the two supports **130a**, **130b** are located at distinct positions and do not come into contact.

(47) In the event of a lateral impact on the chassis, and if the additional tensile stress applied to the elastic cables **120** attached under stress is above the minimum tensile deformation threshold of the cables **120**, the cables **120** are stretched elastically. The support **130a**, **130b** is then driven in a lateral movement in a direction opposite to that of the force applied by inertia effect with respect to

the external structure **110** and then returns to an initial position by spring effect. The impact is thus damped at the support **130a**, **130b** because of the elasticity of the cables **120**. If the additional tensile stress is above the elastic limit of the cables **120**, the cables **120** furthermore suffer plastic deformation making it possible to at least partially absorb the kinetic energy transmitted to the chassis and to damp the impact at the support **130a**, **130b**.

(48) In the case of a vertical impact on the chassis, for example when the drone **100** falls onto the ground, the cables **120** being located between two attachment points **116** in the same direction as that of a force applied to the drone chassis **100**, there is no tensile stress on all the cables **120**. Said cables **120** relax and do not oppose the bending deformation of the external structure **110**. When the external structure **110** comes into contact with the ground, the support **130a**, **130b** is driven, because of its inertia, in a downward vertical movement with respect to the external structure **110**. Such a downward vertical movement of the support **130a**, **130b** generates an overtension on the upper cables **120** attached to the upper attachment point **116** and a relaxation of the lower cables **120** attached to the lower attachment point **116**. At least some of the kinetic energy transmitted to the chassis is converted into elastic energy stored in the cables **120** that have remained under tension, in other words in the upper cables **120**, which makes it possible to reduce the inertia forces at the support **130a**, **130b**. If the impact force is such that the overtension on the upper cables **120** is above the tensile elastic limit of the cables **120**, at least part of the kinetic energy transmitted to the chassis is furthermore absorbed plastically by the cables **120**. The impact is damped at the support **130a**, **130b** but the cables **120** must be replaced for further use of the drone **100**.

(49) FIG. 6 illustrates schematically a view in cross section, along the first cutting plane, of a part of the drone chassis comprising the first arch **111** of the external structure **110** and the set of cables **120** arranged according to a fourth embodiment.

(50) The set of cables **120** connects two supports **130a** and **130b**. A first cable **120** connects a first attachment point **116** of the first arch **111** to the first support **130a**. A pair of two cables connects a second attachment point **116**, opposite to the first attachment point **116** on the first arch **111**, to two points on the second support **130b**. The two supports **130a** and **130b** are furthermore connected together and to a third and fourth attachment point **116** by two groups of three cables **120**, referred to as star-shaped groups. For each star-shaped group, the three cables **120** are connected together by a node N. A first cable **120** in the star-shaped group connects the node N to a point on the first support **130a**. A second cable **120** in the star-shaped group connects the node N to a point on the second support **130b**. A third cable **120** in the star-shaped group connects the node N to an attachment point **116**. The star-shaped groups make it possible to reduce the total length of the cables **120** connecting the supports **130a**, **130b** to an arch **111**.

(51) The third and fourth attachment points **116** form an alignment perpendicular to an alignment formed by the first and second attachment points **116**. Furthermore, the third and fourth attachment points **116** are located in a vertical alignment when the drone **100** is in a stable position.

(52) In order to apply an initial tensile stress to the cables **120** configured according to the fourth embodiment, it suffices to stretch, towards the first attachment point **116**, the first cable **120** connected to the first support **130a**. A tension exerted on the first cable **120** is transferred to the whole of the set of cables **120**.

(53) The cables **120** located between the third and fourth attachment points **116** have a tendency to oppose the bending deformation of the external structure **110** at the time of a lateral impact on the drone **100**.

(54) The cables **120** located between the first and second attachment points **116** have a tendency to oppose the bending deformation of the external structure **100** at the time of a vertical impact on the drone.

(55) FIG. 7 illustrates schematically in plan view a part of the drone chassis comprising the second arch **112** and a set of cables **120** connecting four motor assemblies **700**, each motor assembly comprising a motor and a propeller. A motor assembly **700** is disposed inside the external structure

110. The external structure **110** thus protects the motor assemblies **700** and propeller from damage during an external impact. Furthermore, the external structure **110** makes it possible to avoid a propeller becoming attached on an external object during a flight and makes it possible to protect the environment of the drone **100**.

(56) Each motor assembly **100** is supported by the external structure **110** by means of one or more rigid rods **701** and is furthermore connected to each of the other motor assemblies **700** by a set of cables **120**. The set of cables **120** thus comprises six cables **120** each connecting two motor assemblies **700**. According to one embodiment, the motor assemblies **700** are arranged in a square in one and the same plane and the cables **120** form the sides and the diagonals of the square.

(57) FIG. 8 illustrates schematically in perspective a part of the drone chassis comprising the second horizontal arch **112**, with a set of cables **120** making it possible to hold a support **130** according to a fifth embodiment. The support **130** is connected to six attachment points **116** of the second arch **112** by six distinct cables **120**. Three upper cables **120** are attached to three first attachment points **116** located in the upper part of the second arch **112** and distributed over the circumference of the arch **112**. Three lower cables **120** are attached to three second attachment points **116** located on the second arch **112** below respectively the three first attachment points **116**.

(58) In the event of an impact of the chassis on the ground, for example when the drone **100** falls, the support **130** is driven downwards because of its inertia. The downward vertical movement of the support **130** causes an overtension on the upper cables **120** and a relaxation of the lower cables **120**. Part of the kinetic energy transmitted to the chassis is therefore absorbed or stored elastically by the upper cables **120**, which protects the support **130** by reducing the impact force.

Claims

1. A drone chassis, comprising an external structure that deforms elastically in bending, and furthermore comprising a set of cables connecting at least one support to the external structure, wherein the at least one support is configured to support a functional element of the drone, the set of cables comprising elastic cables that deform elastically in traction under a tensile stress above a threshold, referred to as the natural tensile deformation threshold, the set of cables being attached at several points on the external structure, referred to as attachment points, to hold the at least one support in a stable position with respect to the external structure, wherein the elastic cables are attached under stress by applying, to said elastic cables, an initial tensile stress, and by maintaining the initial tensile stress by attaching to the external structure so that an external force applied to the chassis and generating an additional tensile stress on the elastic cables attached under stress causes a deformation under traction of the cables and a bending deformation of the external structure only if said additional tensile stress exceeds a minimum tensile deformation threshold that is above the natural tensile deformation threshold of the elastic cables.
2. The drone chassis according to claim 1, wherein the set of cables comprises elastic cables made from high-density polyethylene.
3. The drone chassis according to claim 1, wherein a cross section of each elastic cable in the set of cables is determined for reducing an elastic limit of the elastic cables and favouring plastic deformation of said cables when an additional tensile stress applied to the elastic cables attached under stress, and generated by the external force applied to the chassis, is above the minimum tensile deformation threshold of said elastic cables attached under stress.
4. The drone chassis according to claim 1, wherein the external structure comprises two arches, a first arch being located in a plane orthogonal to a second arch, the first arch being in a vertical plane and the second arch being in a horizontal plane when the drone is in a stabilised flight position, and wherein the attachment points to which the at least one support is attached are located on one and the same arch.
5. The drone chassis according to claim 1, wherein the set of cables comprises cables oriented in

several directions.

6. The drone chassis according to claim 1, wherein a first support comprises a flight map, and wherein each cable in the set of cables has a given length for holding the first support comprising the flight map at the centre of the external structure.

7. The drone chassis according to claim 1, wherein each cable in the set of cables has a given length for centring, with respect to the external structure, a centre of gravity of a set of the functional elements supported by the at least one support.

8. The drone chassis according to claim 1, wherein the set of cables connects a plurality of supports to the external structure, each support being connected to the external structure by a subset of the set of cables independently of another support.

9. The drone chassis according to claim 1, wherein the set of cables comprises a subset of cables connecting the at least one support to two attachment points of the external structure, the subset of cables comprising two pairs of cables, each pair of cables connecting one of the two attachment points to the support, and the subset of cables being arranged so that, for each pair of cables, a first cable in the pair of cables connects an attachment point to a first point of the support and a second cable in said pair of cables connects the same attachment point to a second point of the support different from the first point of the support.

10. The drone chassis according to claim 1, wherein the set of cables comprises a subset of cables that connects, between two attachment points, several supports aligned with each other and aligned with said two attachment points.

11. The drone chassis according to claim 4, wherein the set of cables connects the at least one support to six attachment points of the second arch by six distinct cables, the six distinct cables comprising three upper cables connecting the support to respectively three first attachment points, the first attachment points being located at the upper part of the second arch and distributed over the circumference of the arch, and three lower cables connecting the support to respectively three second attachment points, each second attachment point being located on the second arch below a first attachment point.

12. The drone chassis according to claim 1, wherein the chassis supports at least four motor units each comprising a motor and a propeller, each motor unit being attached to the external structure by a rigid rod, and wherein the set of cables comprises a cable connecting each motor unit to each other motor unit.

13. The drone chassis according to claim 1, wherein electric wires are passed inside cables in the set of cables for electrically connecting functional elements together.

14. A drone comprising the drone chassis according to claim 1.
